

The motivation of the paper is to present the applications of LLMs in promoting equity. Most of the discussion around societal impact of LLMs is focused around questions of how to mitigate the biases in LLMs but their role in promoting equity is often ignored. Recent advancements in LLMs mean that the set of possible equity-promoting use cases involving language or text data has significantly increased. The paper presents four promising research directions while keeping risks and cautionary points in view. The authors focus on health as the primary domain for their study, however, all the discussed applications have analogues in non-health domains too.

First equity promoting application of LLMs is the detection of bias improvement. LLMs can improve our ability to detect human bias from text data. Previous research has found evidence of stigmatizing language in clinical notes but has often relied on basic linguistic analysis methods. LLMs can more precisely characterize subtle linguistic biases in tone, sentiments and stereotypes. Despite having biases themselves, they are much better at capture linguistic subtleties compared to keyword search methods. In addition to identifying biases, LLMs can also help mitigate them. They can also be used to accelerate detection of biases in medical textbooks, which would otherwise involve careful analysis from domain experts. Likewise, in law, LLMs could be applied to detect bias in trial transcripts. In hiring, they can determine such biases in letters of recommendation.

Journalists, healthcare workers, activists, and others spend considerable amount of time reviewing relevant literature to curate datasets on various types of discrimination. LLMs have demonstrated impressive performance on extracting structured information from unstructured datasets and thus can be used for all of these works. LLMs can also equity of access to information. In healthcare for instance, for patients who lack easy access to a medical provider or cannot afford to see one, LLMs can provide accurate answers to their queries, and can also simplify the technical jargon to help the patients understand them. It is important to create systems which perform well with multiple languages, and not just English. Chatbots can provide specialized information in other domains too.

LLMs can also improve equity in matching systems as they can interpret complex inclusion and exclusion criteria and suggest eligible individuals from underrepresented groups. In healthcare for instance, making this information available to patients could empower patients to seek out trials they are eligible for. LLMs can also be used to automatically detect violations of equity-driven reporting standards by parsing public documentation for clinical trials, housing or hiring.

The authors also present the steps towards implementation. Firstly, the authors suggest that policy makers should not only try to mitigate equity related harms, but also incentivize equity promoting LLMs. Second recommendation by the authors is to support efforts to ensure responsible and equitable deployments and thorough evaluation of LLMs through basic research funding. The third recommendation by the authors is to consider LLMs as one tool in the toolbox for inequity, rather than a one size fits all kind of solution. LLM enabled innovations must always be coupled with non-AI interventions.

Overall, the paper presents a broad horizon of possibilities for LLM applications in promoting equity. Personally, I have found LLMs to be a great tool for accessing obscure information, which would have otherwise been difficult, if not borderline impossible, to find. Before LLMs were a thing, phrases like "if it is not on the first page of Google, it does not exist" were often joked about, and were responsible for SEO spam from publishers. LLMs are capable of presenting and simplifying information, which is already improving information equity. These capabilities should be responsibly expanded to the various equity related applications as the authors describe.